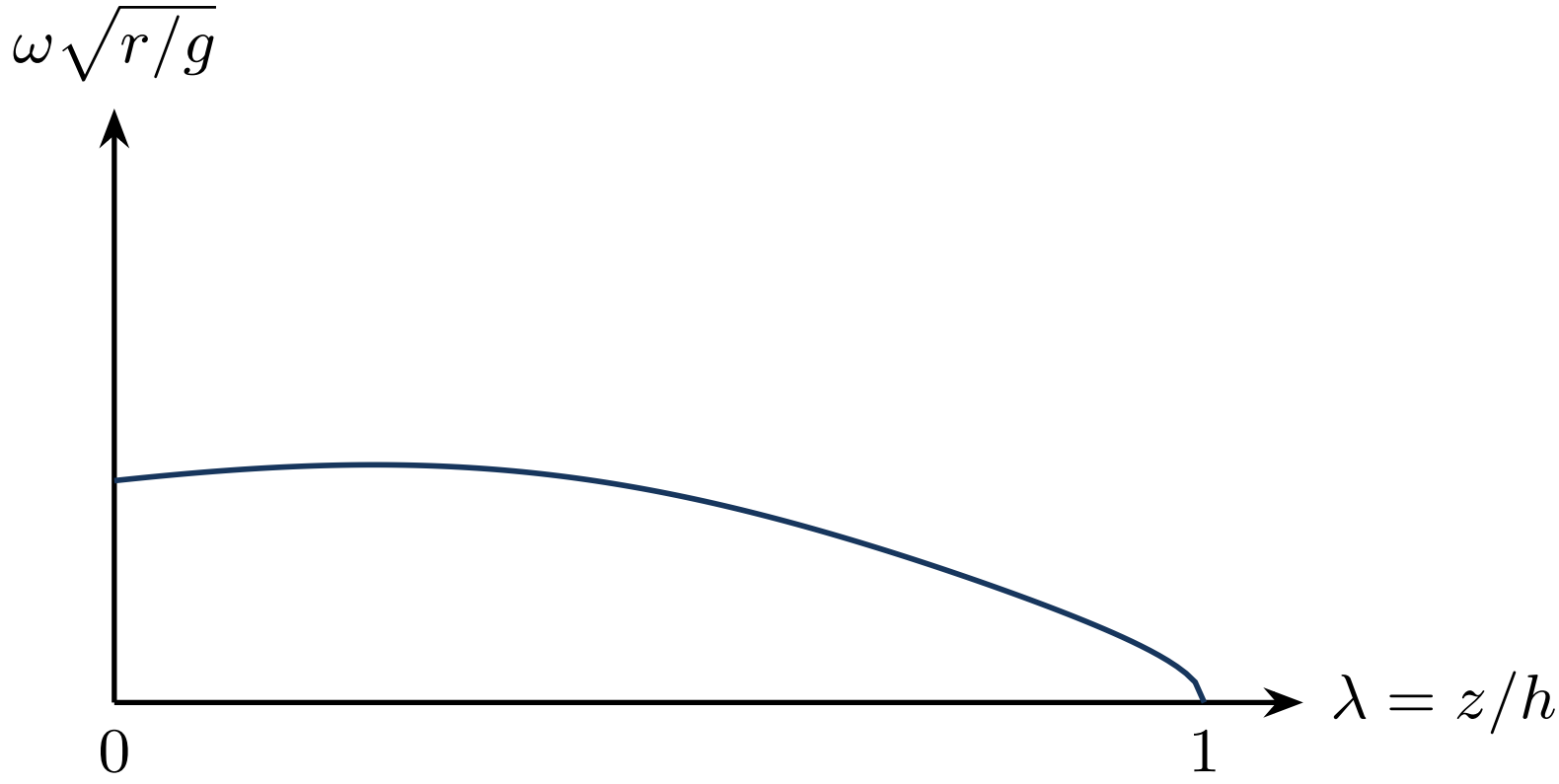

从右向左下滑：上端角速度为零

中点 $\omega^2 = 4\sqrt{3}g/(153r)$，总转角 $\pi/9$